



COMPUTERSHARE COMMUNICATION
SERVICES LIMITED

BMO-US ISO CHECKS

BUSINESS REQUIREMENTS

Author	Chad Trate
Owner	Rachel Hammond
Reviewer	Kashif Mughni
Version	1.41
Date	June 30, 2026

Document History

BMO Document Acceptance

BMO-US Business Representative

Date

CCS Document Acceptance

Matthew Travers
CCS AVP Key Accounts

Date

Kashif Mughni
CCS Solutions Manager

Date

Document History

Owners and List of Contacts

Name	Role	Phone
Kashif Mughni	CCS Solutions Manager	630 568 0232
Anthony Faikus	CCS Solutions Consultant	630 568 0221
Cassandra Bischoff	CCS Business Analyst	312 207 9682
Chad Trate	CCS Business Analyst	612 803 4423

Revision History

Version	Date	Author	Summary of changes	Approved
1.00	May 11, 2021	Anthony Faikus	Initial Release	N/A
1.01	May 19, 2021	Anthony Faikus	Per Feedback from BMO: Added: R22211, R22212, R23220, R22470, R22480, R22482, R22484, R22490, R22492, R22494, R224100R32420, R32425, Appendix B Updated: R21120, R21151, R21211, R22140, R22160, R22420, R22440, R22450, R22460, Appendix A Removed: R21220, R22222, R22290	
1.02	June 24, 2021	Anthony Faikus	Per Feedback from BMO: Added: R21160, R21162	
1.03	August 19, 2021	Anthony Faikus	Per feedback: Added: R22213 Updated: R21140, R21214, R22310, R22320, R22482, R224100, R23110, R23120, R23130, R23140, R32220, R322230, R322240, Appendix A Removed: R21153, R22490, R22492, R22494, R32242	
1.04	October 14, 2021	Anthony Faikus	Per feedback: Updated: R22424, R22434	
1.05	September 21, 2022	Cassandra Bischoff	Per feedback from BMO: Added: R21215, R21216	
1.06	October 19-26, 2022	Charlie Harper	Update: R22470 Added: R22490-R224943	
1.07	November 15, 2022	Cassandra Bischoff	Added: R22285 Updated: R22320	
1.08	February 13, 2023	Cassandra Bischoff	Updated: R22422, Appendix A, Appendix B	
1.09	March 3, 2023	Bryan Rauda	Updated: R23130 Added: R23170	
1.10	March 17, 2023	Cassandra Bischoff	Updated: R21152, R21153, R21154, R21211, R21215, R21216, Appendix B Added: Section 2.1.3 Error Reporting, Appendix D	
1.11	March 30, 2023	Cassandra Bischoff	Updated: Appendix D	
1.12	June 1, 2023	Cassandra Bischoff	Updated: R21215, R23130, Appendix A, and Appendix D Added: R22286	
1.13	June 9, 2023	Cassandra Bischoff	Added: R22214	

Document History

1.14	June 16, 2023	Cassandra Bischoff	Added: R22215
1.15	June 23, 2023	Cassandra Bischoff	Updated: R21153
1.16	June 28, 2023	Cassandra Bischoff	Updated: R21153, R23170
1.17	July 12, 2023	Cassandra Bischoff	Updated: R22482
1.18	August 31, 2023	Cassandra Bischoff	Updated: R22492
1.19	October 25, 2023	Cassandra Bischoff	Added: Section 3.3
1.20	November 7, 2023	Cassandra Bischoff	Updated: R22140, R22310, Appendix A
1.21	November 8, 2023	Cassandra Bischoff	Added: R22140.1
1.22	November 9, 2023	Cassandra Bischoff	Updated: Appendix D
1.23	November 20, 2023	Cassandra Bischoff	Updated: R33110, R33120, R33130, R33140
1.24	November 21, 2023	Cassandra Bischoff	Updated: R33120 Removed: R33140, R33150
1.25	November 28, 2023	Cassandra Bischoff	Updated: R33110, R33120, R33130
1.26	January 18, 2024	Bryan Rauda	[For WR 558] Updated R21140, R23130, R22320, R22321
1.27	February 12, 2024	Bryan Rauda	[For WR576] Updated R22321, R22310, R23130, Appendix A Added R22287, R22288
1.28	March 5, 2024	Bryan Rauda	WR558 Updated R21214
1.29	April 22, 2024	Anna Krocak	WR577. Updated: R22424, R22434
1.30	April 24, 2024	Anna Krocak	WR577. Updated: R22424, R22434, R22444
1.31	April 30, 2024	Anna Krocak	WR599. Updated: R22492
1.32	May 2, 2024	Anna Krocak	WR599. Updated: R22492 Additional requirements for WR599, which remain unchanged: R22470, R22490, R22491, R22493
1.33	May 28, 2024	Cassandra Bischoff	WR612 Updated: R22213 – Bank 29 MICR specs
1.34	June 26, 2024	Cassandra Bischoff	Updated: R21216, R22215
1.35	August 12, 2025	Bryan Rauda	Added R22170
1.36	January 12, 2026	Kashif Mughni	Added R22323 for WR699
1.37	May 28 2026	Chad Trate	Updated R22480 for WR 818, no new positive pay report for reruns
1.38	June 24, 2026	Chad Trate	- R22180 added Option 3 data field mapping - R33012 added Option 3 table example - R33021 added Option 3 table reference - Appendix A added BR.Option3Table to describe added data points
1.39	June 26, 2026	Chad Trate	- R31080 updated with new Ustrd specs - R33012 updated with filename convention

Document History

			- R33021 updated example images and included total bar image
1.40	June 29, 2026	Chad Trate	-Updated R21140 with new file naming convention -Updated R21214 with new acknowledgement naming convention -Updated R31080 with new naming convention -Updated R41030 with new table type -Added R31081 for information section changes
1.41	June 30, 2026	Chad Trate	- removed Option 3 item from R41030 - updated R31080 to reflect ticket number as text

Contents

Table of Contents

1 INTRODUCTION	1
1.1 Purpose of the document	1
1.2 Intended Audience	1
1.3 Project Overview	1
1.4 Business Drivers	1
1.4.1 Improve efficiency	1
1.4.2 Eliminate errors	2
1.5 Scope	2
1.5.1 Dependencies	2
1.5.2 Out of Scope	2
1.6 Reference Documents	2
2 FUNCTIONAL REQUIREMENTS.....	4
2.1 Inbound / Outbound Files	4
2.1.1 File sent from BMO	4
2.1.2 Acknowledgement File sent from Computershare	5
2.1.3 Error Reporting	6
3 COMPOSITION	8
3.1 Data File Details.....	8
3.2 Check Details.....	12
3.3 Document Layout.....	15
3.4 Reporting	21
4 CLIENT ONBOARDING	30
4.1 Standard Client Onboarding Process	30
5 NON-FUNCTIONAL REQUIREMENTS.....	32
5.1 Testing Requirements.....	32
5.2 Production Requirements	32
5.2.1 Production	32
5.2.2 Envelope	32
5.2.3 Lodgement	32
5.2.4 Service Level Agreement	33

Contents

APPENDIX A - ISO PAIN.001 FILE RECORD STRUCTURE..... 35

APPENDIX B - ISO PAIN.002 FILE RECORD STRUCTURE..... 41

APPENDIX C - GLOSSARY OF TERMS 45

1 INTRODUCTION

1.1 Purpose of the document

This document describes the functional requirements for the ISO Check Implementation engagement as requested by BMO-Harris (BMO). The implementation of the functionality contained in this document will result in the production of a Check document for BMO.

The aim of this document is to outline, in business terms, the deliverables for this engagement so that approval can be obtained from the nominated stakeholders.

This document outlines what will be provided by the engagement and not how the requirements will be implemented.

1.2 Intended Audience

The intended audience of this project consists of the following groups:

- BMO Business Representatives
- CCS Account Management Team
- CCS Solutions Team
- CCS Development Team

1.3 Project Overview

BMO US is offering their business clients check printing services where the client's branding information is placed on the checks. BMO will conform to the ISO pain.001.001.03 and pain.002.001.03 XML standards (see Related Documents) as the data to produce and mail the checks. BMO will use CCS to provide the composition, printing, and mailing of checks, as well as providing reconciliation reports for BMO's clients.

1.4 Business Drivers

This section outlines the business drivers for this engagement (i.e. why is the work being undertaken) and is provided to give context for reviewers. The information in this section does not contribute to the requirements of this document.

1.4.1 Improve efficiency

To better understand, improve efficiency, and to stay consistent in the process for check and remittance advice implementation

1.4.2 Eliminate errors

An initiative to create documentation in order to eliminate errors in the composition, printing, mailing, reporting, and reconciliation of Checks

1.5 Scope

1.5.1 Dependencies

The dependent projects for this engagement are:

- BMO will provide all samples for XML files with the corresponding XSD files and data mapping

1.5.2 Out of Scope

The following functionality is considered to be out-of-scope for this solution:

- Including physical inserts/flyers/return envelopes with any Checks
- There will be no color printing for this check implementation (only black/white/grayscale). This includes any logos and images.
- Anything else not explicitly mentioned in this document

1.6 Reference Documents

Document Title	Document
BMO_Comprehensive_Payables_User_Guide	 Comprehensive_Payables_User_Guide.pdf
BMO_pain.001.001.03	 CLIENT - BMO_pain.001.001.03

BMO_pain.002.001.03	 CLIENT BMO_pain.002.001.03
CCS Check Complex Segment	 Check Complex Segment specs.docx
ISO pain.002.001.03 xsd	 pain.002.001.03.xsd

2 FUNCTIONAL REQUIREMENTS

2.1 Inbound / Outbound Files

2.1.1 File sent from BMO

R21110	BMO will provide one XML data file per client, Monday through Friday, to Computershare containing Checks.
R21120	Any file sent on a weekend or holiday, as per NYSE calendar, will be processed in real time, but will not be printed and lodged until the next business day
R21130	XML file will be pre-validated by BMO in accordance to ISO20022 PAIN.001.001.03 format.
R21140	XML file will be named: UAT: T.COMSHARUS.[Customer ID].PAIN001.YYYYMMDD.HHMMSS.NXXXXXX.xml UAT: T.COMSHARUS.[Customer ID].PAIN001.NONVEND.YYYYMMDD.HHMMSS.NXXXXXX.xml UAT: T.COMSHARUS.[Customer ID].PAIN001.VEND.YYYYMMDD.HHMMSS.NXXXXXX.xml UAT: T.COMSHARUS.[Customer ID].PAIN001.CUSTTABLE.YYYYMMDD.HHMMSS.NXXXXXX.xml Prod: P.COMSHARUS.[Customer ID].PAIN001.YYYYMMDD.HHMMSS.NXXXXXX.xml Prod: P.COMSHARUS.[Customer ID].PAIN001.NONVEND.YYYYMMDD.HHMMSS.NXXXXXX.xml Prod: P.COMSHARUS.[Customer ID].PAIN001.VEND.YYYYMMDD.HHMMSS.NXXXXXX.xml Prod: P.COMSHARUS.[Customer ID].PAIN001.CUSTTABLE.YYYYMMDD.HHMMSS.NXXXXXX.xml Where: [Customer ID] = A unique value for that customer within BMO NXXXXXX = A unique serial number assigned by BMO
R21150	Computershare will verify that the XML data file is well-formed.
R21151	If no files are received by 10:00am CT, Computershare will notify BMO of the issue immediately (Email Report R21320)
R21152	If the file was not able to be processed for any of the File Level Validation errors as described in Appendix D, Computershare will contact BMO via email immediately to advise that the file could not be processed. If the file contains any transaction level errors as described in Appendix D, BMO will be advised via the Acknowledgement File (R21211).
R21153	If the file contained duplicate checks (checks whose transit, account and check number combinations are duplicated within the pain 001 file itself or duplicated from a combination of account number and check number from the pain001, and a transit number that was looked up in Computershare's secondary database after bypassing transit number 121100782 being passed in the pain 001 per R22215 within the same pain 001 file itself, or were previously processed in another data file in either of the aforementioned combinations.) Computershare will notify BMO of

	the issue immediately in the Acknowledgement file. The Transaction Status <TxSts> for the duplicate(s) will be set to RJCT. If there are duplicate checks found within the same pain001 input file, Computershare will process the first check in the file, and reject the subsequent duplicate checks. If there is 1 or more RJCT transactions and at least 1 ACCP transaction, the Group Status <GrpSts> will be set to PART. If all transactions are RJCT, the Group Status <GrpSts> will be set to RJCT. Additionally, a CSV detailing the duplicate checks will be generated and sent via sFTP to BMO. A notification email will also be generated. The file will continue processing. All checks that are NOT duplicated will be routed for processing following current processing schedule.
R21154	If during R21153 all items in the file are duplicates the file will cease processing. Document Composition would not be able to be completed if all items were skipped. No report will be generated. A notification email will be generated.
R21160	Computershare will use the record structure as identified in Appendix A.
R21162	If the XML file contains fields that are not identified in Appendix A, Computershare will ignore those fields and will continue file processing.

2.1.2 Acknowledgement File sent from Computershare

R21210	Computershare will create one Acknowledgement Report for every one XML data file received from BMO which will confirm the status of the file.
R21211	Acknowledgement File will be sent to BMO via sFTP in ISO20022 PAIN.002.001.03 with either a RJCT, PART or ACCP status as per ISO20022 specifications. Please refer to Appendix B for the PAIN.002 record structure
R21212	Acknowledgement File will be sent to BMO on a daily basis immediately after the data file has been received and before the Checks have been printed.
R21213	If the XML data file received was invalid, then the Acknowledgement File will contain a validation failure message according to the specifications in the ISO20022 PAIN.002.001.03
R21214	Acknowledgement File will be named: UAT: COMPAYUS.[Customer ID].IPAIN002UST.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml UAT: COMPAYUS.[Customer ID].IPAIN002UST.VEND..YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml UAT: COMPAYUS.[Customer ID].IPAIN002UST.NONVEND.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml UAT: COMPAYUS.[Customer ID].IPAIN002UST.CUSTTABLE.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml

	Prod: COMPAYUS.[Customer ID].IPAIN002USP.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml Prod: COMPAYUS.[Customer ID].IPAIN002USP.VEND.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml Prod: COMPAYUS.[Customer ID].IPAIN002USP.NONVEND.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml Prod: COMPAYUS.[Customer ID].IPAIN002USP.CUSTTABLE.YYYYMMDD.HHMMSS.PAIN002.NXXXXXX.xml Where: [Customer ID] = The unique value for that customer within BMO from the PAIN.001 file NXXXXXX = The unique serial number assigned by BMO from the PAIN.001 file
R21216	Group Status <GrpSts> will be updated as followed: If the File made its way through with no errors found, GrpSts will show ACCP If the File has some but not all rejected checks, GrpSts will show PART If the File has errors in all records, GrpSts will show RJCT If GrpSts shows RJCT, no client reports [zip files] nor any PDF's of the checks will be generated, per the error codes CH21 and NARR as described in Appendix D of this document.

2.1.3 Error Reporting

R21310	The Computershare Account Manager will reach out to BMO via email with file level validation errors as listed in Appendix D.
R21320	An email identifying transaction level error(s) shall be generated if there were any errors in the data file during file processing.
R21330	The subject and body of the "Transaction Level Error(s) Detected" email will be as follows: Subject: "Transaction Level Error(s) Detected" Body of email: PCIPG.NBPD.COMSHARUS.<COMPANY_NAME><Environment>.PAIN001.<NXXXXXX>.xml The file received by Computershare contains one or more errors at the Transaction ID level. Computershare will not process this/these transaction(s) for printing or archiving.

COMMERCIAL IN CONFIDENCE

	Please refer to the Acknowledgement File sent via sFTP in the ISO20022 PAIN.002.001.03 format for the respective for the Reason Reject Codes per Transaction Level failure. This is an automated email. If you are experiencing problems, or you are receiving this in error, please contact Computershare Communication Services and ask for the Client Account Manager. Thank you Where: PCIPG.NBPD.COMSHARUS.<COMPANY_NAME><Environment>.PAIN001.<NXXXXXX>.xml is the input filename from BMO
--	--

3 COMPOSITION

3.1 Data File Details

R31010	Data field specifications will be as per the data field appendix.		
R31020	Data field mapping and various formats will be as per the detailed mapping layout		
R31030	Data formatting rules will be as per the data field appendix		
R31040	The Payee name and mailing address will be as per the following format: <table> <tr> <td> Xpath: /Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ <Cdtr> <Nm>Payee Name</Nm> <PstlAdr> = Postal Address <BldgNb>Payee Building Number</BldgNb> <StrtNm>Payee Street Name</StrtNm> <AdrLine>Address Line</AdrLine> <PstCd>Payee Zip Code</PstCd> <TwnNm>Payee City</TwnNm> <CtrySubDvsn>Payee State</CtrySubDvsn> <Ctry> Payee Country</Ctry> </PstlAdr> </Cdtr> </td><td> e.g. <Nm> <BldgNbr> <StrtNm> <AdrLine> <PstCd> <TwnNm> <CtrySubDvsn> <Ctry> CCS will format the data in the following order: Payee Name Payee Building Number Payee Street Name Payee Address Line City, State, Zip Country </td></tr> </table>	Xpath: /Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ <Cdtr> <Nm>Payee Name</Nm> <PstlAdr> = Postal Address <BldgNb>Payee Building Number</BldgNb> <StrtNm>Payee Street Name</StrtNm> <AdrLine>Address Line</AdrLine> <PstCd>Payee Zip Code</PstCd> <TwnNm>Payee City</TwnNm> <CtrySubDvsn>Payee State</CtrySubDvsn> <Ctry> Payee Country</Ctry> </PstlAdr> </Cdtr>	e.g. <Nm> <BldgNbr> <StrtNm> <AdrLine> <PstCd> <TwnNm> <CtrySubDvsn> <Ctry> CCS will format the data in the following order: Payee Name Payee Building Number Payee Street Name Payee Address Line City, State, Zip Country
Xpath: /Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ <Cdtr> <Nm>Payee Name</Nm> <PstlAdr> = Postal Address <BldgNb>Payee Building Number</BldgNb> <StrtNm>Payee Street Name</StrtNm> <AdrLine>Address Line</AdrLine> <PstCd>Payee Zip Code</PstCd> <TwnNm>Payee City</TwnNm> <CtrySubDvsn>Payee State</CtrySubDvsn> <Ctry> Payee Country</Ctry> </PstlAdr> </Cdtr>	e.g. <Nm> <BldgNbr> <StrtNm> <AdrLine> <PstCd> <TwnNm> <CtrySubDvsn> <Ctry> CCS will format the data in the following order: Payee Name Payee Building Number Payee Street Name Payee Address Line City, State, Zip Country		
R31040 .1	If either just the Payee Street Name data field or both the [optional] Payee Building Number and Payee Street Name data fields are not passed in the PAIN001, then 2 instances of the Payee Address Line data field can be processed and placed on the check as Payee Address Line1 and Payee Address Line2, in the order they are received in the data file. Any additional instances of the Payee Address Line data fields found in the PAIN001 will be ignored, and the file will continue to process. If either just the Payee Street Name data field, or both the Payee Building Number and Payee Street Name data fields are passed in the PAIN001, then only one instance of the Payee Address Line data field can be accepted and printed on the check. Any additional instances of the Payee Address Line data field found in the PAIN001 will be ignored, and the file will continue to process.		

R31041	The Payer name and address will be as per the following format:											
	Xpath: /Document/CstmrCdtTrfInItn/PmtInf/ <Dbtr> <Nm>Payer Name</Nm> <PstlAdr> = Postal Address <<BldgNb>Payer Building Number</BldgNb> <StrtNm>Payer Street Name</StrtNm> <PstCd>Payer Zip Code</PstCd> <TwnNm>Payer City</TwnNm> <CtrySubDvsn>Payer State</CtrySubDvsn> <Ctry> Payer Country</Ctry> </PstlAdr> <CtryOfRes>Payer Country</CtryOfRes> </Dbtr>	e.g. <Nm> BldgNbr> <StrtNm> <PstCd> <TwnNm> <CtrySubDvsn> <Ctry> CCS will format the data in the following order: Payer Name Payer Building Number Payer Street Name City, State, Zip Country										
R31050	BMO will provide the name and address information of their clients in the daily data file.											
R31060	Check Number, Account Number, Routing Number, Check Amount, and Check Date will be included in the data file via the following data fields:											
	<table><tr><td>Check Amount</td><td>/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/Amt/InstdAmt</td></tr><tr><td>Check Number</td><td>/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb</td></tr><tr><td>Check Date</td><td>/Document/CstmrCdtTrfInItn/PmtInf/ReqdExctnDt</td></tr><tr><td>Account Number (Debit Identification)</td><td>Document/CstmrCdtTrfInItn/PmtInf/DbtrAcct/Id/Othr/Id</td></tr><tr><td>Routing Number (Member Identification)</td><td>Document/CstmrCdtTrfInItn/PmtInf/DbtrAgt/FinInstnId/CirSysMmbld/Mmbld</td></tr></table>	Check Amount	/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/Amt/InstdAmt	Check Number	/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb	Check Date	/Document/CstmrCdtTrfInItn/PmtInf/ReqdExctnDt	Account Number (Debit Identification)	Document/CstmrCdtTrfInItn/PmtInf/DbtrAcct/Id/Othr/Id	Routing Number (Member Identification)	Document/CstmrCdtTrfInItn/PmtInf/DbtrAgt/FinInstnId/CirSysMmbld/Mmbld	
Check Amount	/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/Amt/InstdAmt											
Check Number	/Document/CstmrCdtTrfInItn/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb											
Check Date	/Document/CstmrCdtTrfInItn/PmtInf/ReqdExctnDt											
Account Number (Debit Identification)	Document/CstmrCdtTrfInItn/PmtInf/DbtrAcct/Id/Othr/Id											
Routing Number (Member Identification)	Document/CstmrCdtTrfInItn/PmtInf/DbtrAgt/FinInstnId/CirSysMmbld/Mmbld											
R31070	"Additional details on reverse" text will be added below the remittance table when table paginates. Center aligned and ArialBoldMT Size 10											

R31080

Pilgrims Pride Corp will use Table option 3 and come in as:

P.COMSHARUS.[Customer ID].PAIN001.CUSTTABLE.YYYYMMDD.HHMMSS.NXXXXXX.xml:

For accounts using Table Option 3 there is a set of tags

<Ustrd> tags will be processed in pairs. The first is an 13 position semi colon delimited data line
/CstmrCdtTrfIntrn/PmtInf/CdtTrfTxInf/RmtInf/**Ustrd**

	Position 1	Position 2	Position 3	Position 4	Position 5	Position 6	Position 7
	PO	Ticket Number	Contract Price	Quantity Bushels	Gross Amount	Moisture value	Test Weight
Character count	10	16	7	9	10	8	8
Data type	Numeric	Text	Text	Text	Text	Text	Text
Example	1234567890	12345678 90123450	99.9999	9,999.999	99,999.99-	9,999.99	9,999.99

	Position 8	Position 9	Position 10	Position 11	Position 12	Position 13
	Damage number	Foreign Material number	Discount Total	Deferred Pricing	Check Off	Net Amount
Character count	8	8	8	6	6	10
Data type	Text	Text	Text	Text	Text	Text
Example	9,999.99	9,999.99	9,999.99	999.99	999.99	99,999.99-

Second line is a 6 position semi colon delimited data line.

	Position 1	Position 2	Position 3	Position 4	Position 5	Position 6
	PO	Ticket Date	Moisture percent	Test Weight percent	Damage percent	Foreign Material percent
Character count	10	10	5	5	5	5
Data type	Numeric	Text	Text	Text	Text	Text
Example	999	12/12/9999	99.9%	99.9%	99.9%	99.9%

The second line positions should be placed below the first lines columns as follows

Column 1	Column 2	Column 6	Column 7	Column 8	Column 9
Position 1	Position 2	Position 3	Position 4	Position 5	Position 6

There is also a single set of values for the Totals bar at the bottom of the table

<AddtlRmtInf> tag contains a semi colon delimited line
/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Strd/**AddtlRmtInf**

The line has the following values
[Vendor Name];[Material];[Payment Document]

<Nb> tag will store a single instance of the Quantity Bushels Total
/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Strd/RfrdDocInf/**Nb**

<DuePyblAmt> tag will store a single instance of the Gross Amount Total
/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Strd/RfrdDocInf/**DuePyblAmt**

<DscntApldAmt> tag will store a single instance of the Discount Total
/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Strd/RfrdDocInf/**DscntApldAmt**

<CdtNoteAmt> tag will store a single instance of the Net Total
/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/RmtInf/Strd/RfrdDocInf/**CdtNoteAmt**

R31081 Information section at the top of the document for
P.COMSHARUS.[Customer ID].PAIN001.CUSTTABLE.YYYYMMDD.HHMMSS.NXXXXXX.xml:

INFORMATION

Vendor #	123456789ABC000
Vendor	23456789012 NAME D. PERSON
Material	CORN #2 YELLOW 123456
Payment Document	1234567890
Date	06/10/2026
Check No.	000001079

Where:

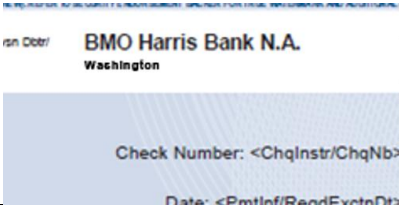
Value	Field
-------	-------

	Vendor #	<Cdtr/Id/OrgId/Othr/Id>
	Vendor	<PmtInf/CdtTrfTxInf/RmtInf/S trd/AddtlRmtInf> <AddtlRmtInf> is a semi colon delimited line. Vendor name is the first field.
	Material	<PmtInf/CdtTrfTxInf/RmtInf/S trd/AddtlRmtInf> <AddtlRmtInf> is a semi colon delimited line. Material is the second field.
	Payment Document	<PmtInf/CdtTrfTxInf/RmtInf/S trd/AddtlRmtInf> <AddtlRmtInf> is a semi colon delimited line. Payment Document is the third field.
	Date	<PmtInf/ReqdExctnDt>
	Check No.	<ChqInstr/ChqNb>

3.2 Check Details

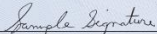
R32010	Check MICR line will be in MICR ink using OCA font using Gauge 56.
R32011	If the routing number is '071000288', then CCS will apply the MICR as per the following MICR spec:  Bank 28 MICR Spec.pdf
R32012	If the routing number is '071915580', then CCS will apply the MICR as per the following MICR spec:

	 Commercial CDA B2B Mic Specs.pdf
R32013	If the routing number is '071025661', then CCS will apply the MICR as per the following MICR spec:  MICR Specifications - Checks - Bank 29 (
R32014	If the routing number is '125107888', then CCS will apply the MICR as per the following MICR spec:  Commercial CDA Bank28 (BotW) MICR
R32015	If the routing number passed in the Pain001 file is 121100782, Computershare will not add routing number 121100782 to the check, and instead lookup the account number in Computershare's Encompass2 database to access the routing # to be printed on the check. If the lookup for the account number in Computershare's Encompass2 database doesn't return any mapped routing number, the check will be rejected. Rejection will be per the Business Condition for "Transit [Routing] Numbers/ MICR Details invalid" listed in Appendix D of this document.
R32020	Bank fraction for BMO Checks will be as per BMO MICR specs
R32030	BMO name and address to be printed as follows:
R32040	BMO logo to be printed in grayscale
R32050	A single account number will be linked to one specific signature and logo.
R32060	BMO will provide a client logo.
R32061	Logo will be printed in black and white / grayscale.
R32062	If a client logo is not provided, logo will not print.
R32063	Computershare (CCS) will require signoff from BMO if a logo file needs to be changed or updated.
R32070	Default font for all Check document text will be Arial 10pt.
R32080	The memo content area created will be able to fit a maximum of 80 characters.

R32081	The memo content will always be preceded by the label "Memo: "
R32082	The memo will always have an 80-character underline (even when there is no memo content being populated).
R32083	The 80-character memo underline will wrap to the second line.
R32084	The memo content may wrap to the second line depending on the length of the content
R32085	If there is a second occurrence of the memo field in the data file, Computershare (CCS) will concatenate the two fields by adding a space between them.
R32086	If there are any more than 2 occurrences of the memo field in the data file, Computershare (CCS) will disregard any additional occurrences, as outlined in the pain.001 xsd: <xs:element name="MemoFld" type="Max35Text" minOccurs="0" maxOccurs="2"/>
R32087	For accounts that require a void text, example Void after 60 days, it will be placed In the top middle portion of the check please see screengrab. If an account produces cashier's checks only, then the text will only say: CASHIER'S CHECK 
R32088	For accounts that require the property state to be added instead of Chicago, Illinois, it will be placed under BMO Harris Bank in the check portion. See screengrab below 

Computershare will create the check document as per the following layout, data mappings, and data client file:

If client file is P.COMSHARUS.[Customer ID].PAIN001.YYYYMMDD.HHMMSS.NXXXXXX.xml, then the check layout will look like the below.

00CDP0066 / R		00000A	GROUP STREAM ENV CONTRACT RAND GEN NUMS_RUNSET# SHEET		
WARNING: MULTIPLE EARTY FEATURES. THE FACT OF THIS CHECK HAS A BLUE BACKGROUND AND FLUORESCENT INK (HOLD UNDER BLACKLIGHT TO VIEW). REFER TO SECURITY ENDORSEMENT BACKFOR FOR THE WATERMARK AND ADDITIONAL FEATURES.					
		Die:PstAdr/Bldg Cntr:PstAdr/TwnNm Cntr:PstAdr/TwnNm, Cntr:PstAdr/CitySubDivns Clear: PstAdr/PstCd		BMO Harris Bank N.A. Chicago, Illinois	2-28 70
Pay	\$ <Amt/InstdAmt>	Check Number: <ChqInstr/ChqNb>			
To the order of	<Cdnr/Nm> <Cdnr/PstlAdr/BldgNb> <Cdnr/PstlAdr/StNrNm> <Cdnr/PstlAdr/AptLine> <Cdnr/PstlAdr/TwnNm> <Cdnr/PstlAdr/CitySubDivns> <Cdnr/PstlAdr/PstCd>	Date: <PmtInfr/ReqdExctnDt>			
		\$ <Amt/InstdAmt>			
 Security Features Details on Back		 Authorized Signature(s)			

If client file is P.COMSHARUS.[Customer ID].PAIN001.VEND.YYYYMMDD.HHMMSS.NXXXXXX.xml, then the check layout will look like the below.

COMMERCIAL IN CONFIDENCE



ABC COMPANY

Doc: PstAdr/BldgNb Doc: PstAdr/CtryNm
Doc: PstAdr/TwnNm Doc: PstAdr/CtrySubDvsn Doc: PstAdr/PstCd

000001

|||||

<CdrNm>
<Cdr/PstAdr/BldgNb>
<Cdr/PstAdr/StrtNm>
<Cdr/PstAdr/AdtLine>
<Cdr/PstAdr/TwnNm>
<Cdr/PstAdr/CtrySubDvsn>
<Cdr/PstAdr/PstCd>

BMO Data Mockup

VENDOR

Page 1 of 1

INFORMATION

Vendor	<CdrId/OrgId/CtrId>
Date	<PmtIn/ReqdExtnDt>
Check Number	<ChqInstr/ChqNb>

ACCOUNT DETAILS

INVOICE DATE	CLAIM NUMBER / INVOICE NUMBER	INVOICE AMOUNT	DEDUCTIONS	INVOICE NET AMOUNT
<RmtIn/Strd/Rfrd DocIn/Rfrd>	<RmtIn/Strd/Rfrd DocIn/ Nb>	<RmtIn/Strd/Rfrd DocAmt/ DuePblyAmt>	<RmtIn/Strd/Rfrd DocAmt/ DsontApdlyAmt>	<RmtIn/Strd/Rfrd DocAmt/ CdtNoteAmt>
<RmtIn/Strd/AdtRmtIn>				
<RmtIn/Strd/Rfrd DocIn/Rfrd>	<RmtIn/Strd/Rfrd DocIn/ Nb>	<RmtIn/Strd/Rfrd DocAmt/ DuePblyAmt>	<RmtIn/Strd/Rfrd DocAmt/ DsontApdlyAmt>	<RmtIn/Strd/Rfrd DocAmt/ CdtNoteAmt>
<RmtIn/Strd/AdtRmtIn>				

<RmtIn/Ustrd> Additional details on reverse

PLEASE DETACH BEFORE DEPOSITING

003COP0299 / R

00000A

GROUP STREAM ENV CONTRACT RAND GEN NUUS RUN SET SHEET



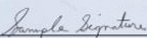
ABC COMPANY

Doc: PstAdr/BldgNb Doc: PstAdr/CtryNm
Doc: PstAdr/TwnNm Doc: PstAdr/CtrySubDvsn Doc: PstAdr/PstCd

BMO Harris Bank N.A.

Chicago, Illinois

2-28
70

Pay	\$<Amt/InstdAmt>	Check Number: <ChqInstr/ChqNb>
To the order of	<CdrNm> <Cdr/PstAdr/BldgNb> <Cdr/PstAdr/StrtNm> <Cdr/PstAdr/AdtLine> <Cdr/PstAdr/TwnNm> <Cdr/PstAdr/CtrySubDvsn> <Cdr/PstAdr/PstCd>	Date: <PmtIn/ReqdExtnDt> \$ <Amt/InstdAmt>  Authorized Signature(s)

 Security Feature Details on Back

⑈0000000000⑈ ⑈0011111788⑈ 335 987 0991⑈

COMMERCIAL IN CONFIDENCE

--	--

If client file is P.COMSHARUS.[Customer ID].PAIN001.NONVEND.YYYYMMDD.HHMMSS.NXXXXXX.xml, then the check layout will look like the below.

For any Holland accounts or for accounts that require the use of Table type option 2 then the check layout will look like the below

[illegible]

R33012

If client file is P.COMSHARUS.[Customer ID].PAIN001.CUSTTABLE.YYYYMMDD.HHMMSS.NXXXXXX.xml, then the check layout will look like the below.

For any Pilgrim's Pride Corp accounts or for accounts that require the use of Table type option 3 then the check layout will look like the below

PILGRIM'S PRIDE CORPORATION TRUCK DELIVERY SETTLEMENT SHEET

PO/Line	Ticket Date	Contract Price	Quantity Bushels	Gross Amount	Moisture	Test Weight	Discounts Damage	Foreign Material	Total	Fees Check Off	Deferred Pricing	Net Amount
1234567890	33335 06/12/2026	4.8800	372.143	1,816.06	0.00 13.1	0.00 59.8	0.00 0.3%	0.00 0.2%	0.00	0.00	0.00	1,816.05
2222222222	7897897897 987654 12/12/2026	12.34567	1234.56789	1,234,567.00	000000.00 000.00	000000.00 000.00	000000.00 000.00%	000000.00 000.00%	112,345.67	2,345.67	2,345.67	1,234,567.00

The table will be mapped with following data line R31080

R33020

In the Account Details table, Computershare will account for an optional open text line for each invoice. Text will be in the <AddtlRmtlnf> tag of the PAIN.001 file and will be printed to extend across the columns of the table. Each occurrence of <AddtlRmtlnf> is to appear in a new line below the previous line.

ACCOUNT DETAILS

INVOICE DATE	INVOICE NUMBER	INVOICE AMOUNT	DISCOUNT AMOUNT	INVOICE NET AMOUNT
02/23/2021	3718	\$819.34	\$0.00	\$819.34
DO NOT PRINT NOTE//				
02/30/2021	3719	\$0.00	\$0.00	\$0.00
INVOICE PAID				

Unstructured below remit table

R33021

The remittance table labels will be different based on client data file name

If client data file is P.COMSHARUS.[Customer ID].PAIN001.YYYYMMDD.HHMMSS.NXXXXXX.xml then the labels in the remittance table will produce the below

ACCOUNT DETAILS

INVOICE DATE	INVOICE NUMBER	INVOICE AMOUNT	DISCOUNT AMOUNT	INVOICE NET AMOUNT
--------------	----------------	----------------	-----------------	--------------------

If client data file is P.COMSHARUS.[Customer ID].PAIN001.VEND.YYYYMMDD.HHMMSS.NXXXXXX.xml then the labels in the remittance table will produce the below

ACCOUNT DETAILS

INVOICE DATE	CLAIM NUMBER / INVOICE NUMBER	INVOICE AMOUNT	DEDUCTIONS	INVOICE NET AMOUNT
--------------	-------------------------------	----------------	------------	--------------------

If client data file is P.COMSHARUS.[Customer ID].PAIN001.NONVEND.YYYYMMDD.HHMMSS.NXXXXXX.xml then the labels in the remittance table will produce the below

ACCOUNT DETAILS

DATE OF LOSS	CLAIM NUMBER	PAID AMOUNT	DEDUCTIONS	NET PAID AMOUNT
--------------	--------------	-------------	------------	-----------------

For any Holland accounts or for accounts that require the use of Table type option 2 then the labels in the remittance table will produce the below

ACCOUNT DETAILS

INVOICE NUMBER	REFERENCE TYPE	REFERENCE ID	EFFECTIVE DATE	TOTAL CURRENT AMOUNT	DISCOUNT AMOUNT	INVOICE ADJUSTMENT
----------------	----------------	--------------	----------------	----------------------	-----------------	--------------------

For any Pilgrim's Pride Corp accounts or for accounts that require the use of Table type option 3 then the labels in the remittance table will produce the below

PO/Line	Ticket Date	Contract Price	Quantity Bushels	Gross Amount	Moisture	Test Weight	Discounts	Foreign Material	Total	Fees	Net Amount
							Damage			Check Off	
										Deferred Pricing	

With a total bar at the end with the following

	Total: 1234.56789 1,234,567.00			112,345.67	1,234,567.00
	Total Amount: 96,748.46				
	Payment Document: 1234567890				Amount: 96,748.46
R33022	All field labels will be printed in English only				
R33023	No remittance table will be printed if the check is a CASHIER'S CHECK, per R22287				

3.4 Reporting

R34010	Computershare will create the following client-level reports for each client Pain001 input data file: - Disbursement Summary Report - Check Register Report - Exception Summary Report																											
R34020	Disbursement Summary Report will be generated per each Pain001 input data file to report the count and value of following items, which group by bank account number initiated the transaction: - Checks to be printed - Total Checks Issued - Checks Printed - Checks Rejected																											
R34022	Disbursement Summary Report will be named using the following convention: Dis_summ_BMO<FILE_NAME>_<BANK>_<CURRENCY>.xls Where: <FILE_NAME> is the full name of the original input file <BANK> is the name of the financial institution processes the transaction (i.e. BMO) <CURRENCY> is the currency code of the report (i.e. CAD or USD)																											
R34024	<table><tr><td colspan="3">The Disbursement Summary Report will be created with the following format:</td></tr><tr><td colspan="3">Computershare</td></tr><tr><td colspan="3">Disbursement Summary Report</td></tr><tr><td colspan="3"><Company Name></td></tr><tr><td colspan="3"><Original Input File Name></td></tr><tr><td colspan="3">DATE: <Earliest date of all Checks></td></tr><tr><td></td><td></td><td></td></tr><tr><td>INPUT TOTALS</td><td>Count</td><td>Value</td></tr><tr><td>CHECKS TO BE PRINTED</td><td><count of checks to be printed></td><td><value of checks to be printed></td></tr></table>	The Disbursement Summary Report will be created with the following format:			Computershare			Disbursement Summary Report			<Company Name>			<Original Input File Name>			DATE: <Earliest date of all Checks>						INPUT TOTALS	Count	Value	CHECKS TO BE PRINTED	<count of checks to be printed>	<value of checks to be printed>
The Disbursement Summary Report will be created with the following format:																												
Computershare																												
Disbursement Summary Report																												
<Company Name>																												
<Original Input File Name>																												
DATE: <Earliest date of all Checks>																												
INPUT TOTALS	Count	Value																										
CHECKS TO BE PRINTED	<count of checks to be printed>	<value of checks to be printed>																										

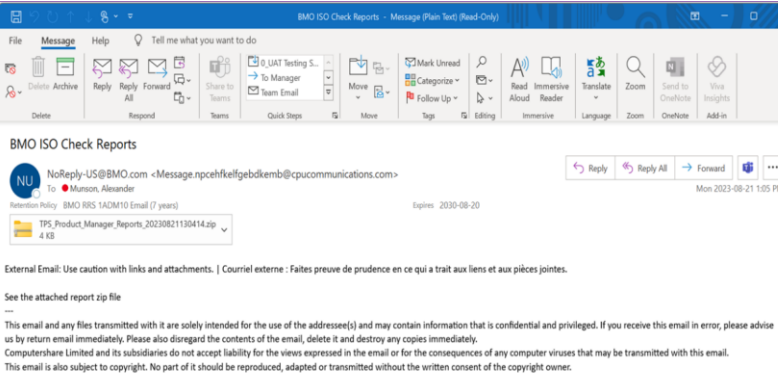
		TOTAL ISSUED	<total count of checks actually printed>	<total value of checks actually printed>
		RUN TOTALS	Count	Value
		CHECKS PRINTED - <bank account number>	<count of checks printed for a specific bank account>	<value of checks printed for a specific bank account>
		CHECKS REJECTED - <bank account number>	<count of checks rejected for a specific bank account>	<value of checks rejected for a specific bank account>
		TOTAL (<Currency>)	<total count of checks to be printed>	<total value of checks to be printed>
Note A: Bank Account_Number will appear with the first four digits masked with 'XXXX' Note B: Values in the "Count" column will be formatted as "N,NNN" Note C: Values in the "Value" column will be formatted as "\$N,NNN.nn" NOTE D: The Run Totals - Checks Printed and Checks Rejected will repeat in every account number in the Pain001.xml An example of this report is in the below attachment:  Disbursement Summary Report EX/				
R34030	Check Register Report will be generated after each Pain001 input data file is processed to report the following items. For each bank account number involved in the Pain001 input data file , one single Check Register Report will be generated. Therefore, each Pain001 input data file may have multiple Check Register Reports. Each report will contain the following information: • Check Number / Check/ • Payee Name • Check Amount • Total Count of Check • Total Amount			
R34032	Check register Report will be named using the following convention: Chq_register_BMO<FILE_NAME>_<sequence>_<Account Number>.xls Where: <FILE_NAME> is the full name of the original input file <Account Number> is the account where the fund flow outs, the first 4 digits of account number should be replaced by 'x'			
R34034	The Check Register will be created with the following format:			

	Computershare															
	Check Register Report															
	<Company Name>															
	<Orginal Input File Name>															
	DATE: <Earliest Date of all Checks >															
	ACCOUNT NUMBER: <Bank Account Number BMO client, first 4 digit masked>															
	CHECK NUMBER	DATE	PAYEE NAME	CHECK AMOUNT												
	<Check Number >	<Check Date >	<Payee name>	<Amount>												
	<total count of checks actually printed >		<total value of checks actually printed>													
COUNT:		TOTAL:														
Note A: Values in the "CHECK AMOUNT" column will be formatted as "\$N,NNN.nn"																
An example of this report is in the below attachment:																
																
Check Register Report EXAMPLE.xls																
R34040	An Exception Summary Report will be generated after each Pain001 input data file is processed to provide a summary view of ALL checks that cannot be printed, including reason. The report contains the information of - o Run Date - o Run # - o File Name - o Originator Account Number - o Check Number - o Reason															
R34042	The Exception Summary Report will be named using the following convention: Exception_<Input File Name>.xls Where: <Input File Name> is the full name of the original input file used processed by the run															
R34044	The Exception Summary Report will be created with the following format: <table><tr><td colspan="4">Computershare</td></tr><tr><td colspan="4">Exception Summary Report</td></tr><tr><td colspan="4"><Company Name></td></tr></table>				Computershare				Exception Summary Report				<Company Name>			
Computershare																
Exception Summary Report																
<Company Name>																

	<Original Input File Name>		
	DATE: <Run Date>		
	ACCOUNT	CHECK NUM	EXCEPTION
	<Account Number>	<Check Number>	<Exception Detail>
	Note A: In cases where no checks meet the criteria for exception, the report should include a statement: 'No exceptions were found.' In the ACCOUNT and EXCEPTION column. An example of this report is in the below attachment:  Exception Summary Report EXAMPLE.xls		
R34050	All client-level reports for each client will be enclosed in a zip file and will be emailed to the client email address The contents of the zip file will be as illustrated below: ➤ Pain001 input data file has only a single bank account number ○ <i>disb_summ_BMOISO-CCS-WENERCAN-230316738618_BMO_USD.xls</i> ○ <i>chq_register_BMOISO-CCS-WENERCAN-230316738618_1_111.xls</i> ○ <i>exception_BMOISO-CCS-WENERCAN-230316738618.xls</i> ➤ Pain001 input data file has multiple bank account numbers (e.g. 2 distinct 7-digit account numbers ending with "111" and "222") ○ <i>disb_summ_BMOISO-CCS-WENERCAN-230316738618_BMO_USD.xls</i> ○ <i>chq_register_BMOISO-CCS-WENERCAN-230316738618_1_111.xls</i> ○ <i>chq_register_BMOISO-CCS-WENERCAN-230316738618_2_222.xls</i> ○ <i>exception_BMOISO-CCS-WENERCAN-230316738618.xls</i> NOTE: for each Check Register Report excel file, the first 4 digits of account number should be replaced by 'x' in respective file names.		
R34060	All client-level report emails sent to BMO's clients will be from a BMO domain and will be embedded with "No Reply-US" Note: This is a PROD only requirement, and hence client level report emails are not generated in UAT environment. An example of this mail format is illustrated in the below screenshot, and the attached zip file in the screenshot is as referred in requirement R22450.		

Commented [AK1]: Added more detailed description per BMO's request

COMMERCIAL IN CONFIDENCE

	 BMO ISO Check Reports To: NoReply-US@BMO.com <Message.npcehfkefgebdkemb@cpucommunications.com> Retention Policy: BMO RMS 1ACM10 Email (7 years) Expires: 2030-09-20 TPS_Product_Manager_Reports_20230821130414.zip 4 KB External Email: Use caution with links and attachments. Courriel externe : Faites preuve de prudence en ce qui a trait aux liens et aux pièces jointes. See the attached report zip file --- This email and any files transmitted with it are solely intended for the use of the addressee(s) and may contain information that is confidential and privileged. If you receive this email in error, please advise us by return email immediately. Please also disregard the contents of the email, delete it and destroy any copies immediately. Computershare Limited and its subsidiaries do not accept liability for the views expressed in the email or for the consequences of any computer viruses that may be transmitted with this email. This email is also subject to copyright. No part of it should be reproduced, adapted or transmitted without the written consent of the copyright owner.
R32070	Computershare will create the following reports and transmit directly to BMO: • Positive Pay Report • Monthly Billing Report
R32080	If a check is being produced by the contract, a Positive Pay Report must be created.
	If a file is rerun after a Positive Pay Report has been created and delivered, no new Positive Pay Report will be created

R22482

The positive pay report will be named:

•

If xpath: CCS/Pack/Check_Data/Routing_Number is '071000288' or '07195580' or '125107888', the file name will be:

○

UAT COMPAYUS.CCSUSTEST.IDIRECTCK.YYYYMMDDHHMMSS.NXXXXXX.ARP530R.txt

•

Prod: COMPAYUS.CCSUS.IDIRECTCK.YYYYMMDDHHMMSS.NXXXXXX.ARP50T8.txt

•

If xpath: CCS/Pack/Check_Data/Routing_Number is '071025661', the file name will be:

○

UAT:
COMPAYUS.CCSUS29TEST.IDIRECTCK.YYYYMMDD.HHMMSS.NXXXXXX.ARP530S.txt

○

Prod: COMPAYUS.CCSUS29.IDIRECTCK.YYYYMMDD.HHMMSS.NXXXXXX.ARP52ZW.txt

NOTE: The date and timestamp in the filename will be the date and time from the input file in xpath: CCS/Pack/File_Information/File_Name.

And "NXXXXXX" represents a unique serial number assigned by BMO in the input file

R22484

The BMO Positive Pay Report will be a fixed-length file with the following data structure:

Detail Record

Record Size

194

Record Format

Fixed block

Numeric Fields

Right justify and zero fill

001	001-001	1	Constant Value = C
002	002-004	3	Bank Number *
003	005-006	2	Filler - zeros
004	007-016	10	Account Number*
005	017-017	1	Filler - blank/space
006	018-018	1	Transaction Type: R=Register, V = Void / Cancel
007	019-019	1	Action Indicator: A = Add, D = Delete
008	020-020	1	Filler - blank/space
009	021-030	10	Check Serial Number

	010	031-040	10	Check Amount
	011	041-046	6	Issue Date (MMDDYY)
	012	047-066	2	User Information
	013	067-194	12	Payee**

(*) Bank Number and Account Number should be verified with BMO Harris Bank Implementation Specialist.

(**) Payee Name field is left justified. Field can truncate at the end of the payee name, there is no requirement to fill to maximum field length.

Using the variable data inputs in record positions 018 and 019, the standard check issue file format can cover the four following types of check issue activities:

RA	Register Add	Adds item as an outstanding issued check
RD	Register Delete	Removes the item as an outstanding issued check
VA	Void Add	Indicates that an outstanding item is now "voided/cancelled" and should not be paid.
VD	Void Delete	Used to delete the "void/cancel" status on an issued item. Item becomes a valid issued check once again and will be processed by the system if presented.

R22490	A Monthly Billing Report will be created on the first of the month, capturing specific job details from the entire previous month.		
R22491	The Monthly Billing Reports will have the following naming convention: COMPAY.BILLCCSUS.IDP_XAA_vvvymmddhhmmss.txt		
R22492	The Monthly Billing Report will be a fixed-length file with the following data structure, consisting of a Header row, Records row, and Trailer row: Header Row		
	Position	Field Name	Description
	1-3	Record Type	Hardcoded as 'HDR'
	4-7	Line of Business Name	Hardcoded as 'TPS' Text is right justified
	8-10	System ID	Hardcoded as 'UNX'

11-20	Job Name	Hardcoded as 'CCSUSCKBIL'
21-50	Table Name	Hardcoded as 'NO_TABLE' Text is right justified
51-58	Data Date	Last calendar date of the queried month Format: 'YYYYMMDD'
59-66	Run Date	Date that the Billing Report was run Format: 'YYYYMMDD'
67-74	Run Time	Time at which the Billing Report was run Format: 'HH:MM:SS'
75-76	Generation	Hardcoded as '01'
7-84	Next Data Date	Date on which the next Monthly Billing Report will be available Format: 'YYYYMMDD'
85-90	Time zone	Hardcoded as 'CT' Left justify
Record Rows		
Position	Field Name	Description
<u>1-3</u>	<u>REC_TYP</u>	"BDY"
<u>4-7</u>	<u>SUR-BANK_NUMBER</u>	Hardcoded as <u>UAT = '613'</u> <u>Prod = '795'</u>
<u>8</u>	<u>SUR-APPLICATION-ID</u>	Hardcoded as 'D'
<u>9-28</u>	<u>SUR-ACCOUNT-NUMBER</u>	Value from input file <u>/Document/CstmrCdtTrfIntrn/PmtInf/DbtrAcct/Id/Othr/Id</u> Zero filled to 20 characters
<u>29</u>	<u>SUR-DEPOSITORY-CATEGORY</u>	Empty
<u>30-32</u>	<u>SUR-RECORD-TYPE-020</u>	Hardcoded as '020'
<u>33-42</u>	<u>SUR-EFFECTIVE-DATE</u>	Last calendar date of the queried month Format: 'YYYY-MM-DD'

<u>43-52</u>	<u>SUR-POSTING-DATE</u>	Input file's date within file name P.COMSHARUS.[Customer ID].PAIN001.YYYYMMDD.HHMMSS.NXXXXXX.xml Format: 'YYYY-MM-DD'
<u>53-55</u>	<u>SUR-SOURCE</u>	Hardcoded as '860'
<u>56-63</u>	<u>SUR-CONTROL</u>	Empty
<u>64-68</u>	<u>SUR-SERVICE-CODE</u>	70203 will indicate that the SUR-UNITS value is the volume of checks 70204 will indicate that the SUR-UNITS value is the number of additional images Each Service Code will be separate and consecutive rows in the Monthly Billing Report
<u>69</u>	<u>SUR-UNITS-IND</u>	Hardcoded as '+'
<u>70-76</u>	<u>SUR-UNITS</u>	Number of units of the SUR-SERVICE-CODE Zero-filled
<u>77-111</u>	<u>SUR-DESCRIPTION</u>	Empty
<u>112-146</u>	<u>SUR-ADD-DESC</u>	Empty
<u>147-151</u>	<u>FILLER</u>	Empty
TRAILER Row		
<u>Position</u>	<u>Field Name</u>	<u>Description</u>
<u>1-3</u>	<u>Record Type</u>	Hardcoded as 'TRL'
<u>4-11</u>	<u>Run Date</u>	Date that the Billing Report was run Format: 'YYYYMMDD'
<u>12-22</u>	<u>Record Count</u>	Number of Record rows in the Monthly Billing Report. Header and Trailer rows to be excluded Zero-filled
<u>23-44</u>	<u>Checksum1</u>	Empty
<u>45-66</u>	<u>Checksum2</u>	Empty
<u>67-88</u>	<u>Checksum3</u>	Empty
R22493	CCS will transmit the Monthly Billing Report to BMO via the outbound sFTP on the first of each month by 8:30AM CST	
R224100	CCS will transmit the Positive Pay Report to BMO via the outbound sFTP	

4 CLIENT ONBOARDING

4.1 Standard Client Onboarding Process

R41010	BMO clients will be onboarded to the solution using a client onboarding excel form.
R41020	Computershare will provide BMO with the excel document template to enter the onboarding information for each client
R41030	The onboarding form will capture the following data: • Company Name • Account Number • Special Handling Address – Name • Special Handling Address – Attn: • Special Handling Address • Business Rules for using a signature • Courier Option (Optional) • Courier Account Number (Required if courier option is populated) • Courier Bill to Name (Required if courier option is populated) • Courier Bill to Address (Required if courier option is populated) • Courier Bill to Email Address (Required if courier option is populated) • Company Email Address • Routing Number (Optional) ◦ Drop down list with the following options (only one option can be selected): ▪ 071000288 ▪ 071025661 • Customization (Optional) ◦ Void text ▪ This will be a free form text. If the account produces cashier's checks only, then this value needs to be: CASHIER'S CHECK • Bank Address ◦ Bank Address Line 1 is set in the Aardvark Autoproc Extra Argument ◦ Bank Address Line 2: [Freeform Text Field] ◦ Bank Address Line 3: [Freeform Text Field] ◦ Bank Address Line 4: [Freeform Text Field] - If bank address 1, 2, 3 and 4 are empty we will use the bank details • Table type (dropdown list) ◦ Option 1- default

	○ Option 2- 7 column table ○ Option 3- Pilgrim delimited Table • Zip file password for report
R41040	Computershare will provide BMO with an extract of the client lookup tables after processing the client onboarding form. NOTE: Computershare will omit the value contained in <Zip_Password> from the extract.
R41050	Extract file will have the following naming convention: BMO_Client_Lookup_Data_YYYYMMDD.xlsx
R41060	Computershare will send the extract file to BMO via sFTP
R41070	In the PAIN.001 file there is a special handling field <Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ChqInstr/DlvryMtd/Prtry>. If the value is blank or the tag is not present, then the pack is sent via USPS to the payee. If the value is '01', it should be couriered back to the debtor. If the value is '02' then it is couriered to the payee. If it is '03', then it is couriered to the special handling address. These '01', '02', and '03' values are set up in by account number Computershare's database with information about the respective courier accounts to be used. If the account number does not have the courier account information stored in the database, the Pain 001 will fail to process, and BMO will be notified immediately via the Computershare Account Manager.

5 NON-FUNCTIONAL REQUIREMENTS

5.1 Testing Requirements

R51010	BMO will contact their designated Computershare Account Manager to send test files to Computershare's UAT environment in order to see PDF samples.
R51020	BMO will supply test files for the Check layout to allow Computershare to perform User Acceptance Testing.

5.2 Production Requirements

5.2.1 Production

R52110	Content from Checks may need to be printed on more than one sheet of paper depending on the length of the content.
R52120	Checks will be printed on generic Computershare check stock.
R52121	Check perforation will be at 3 1/3 inches from the bottom of the page.
R52140	Content that needs to flow beyond the perforation mark will flow to the front of the next sheet and be printed on plain white letter size paper (8.5in x 11in).

5.2.2 Envelope

R52210	Checks will be inserted into the Computershare generic large window envelopes
R52211	Depending on the number of sheets of paper for each document, a #10, 6x9 or 9x12 envelope may be used.
R52220	Large window envelope #10 will be used for a total page sheet count maximum of 1 to 5 sheets.
R52230	Large window envelope "6x9" will be used for a total page sheet count maximum of 6 to 9 sheets.
R52240	Large window envelope "9x12" will be used for total page sheet count maximum of 10 or more sheets.
R52241	The maximum number of sheets of paper for one document will not exceed the capacity of a 9x12 envelope.

5.2.3 Lodgement

R52310	Printing will occur Monday to Friday during regular business hours excluding weekends, non-working days, and holidays as per the NYSE calendar.
---------------	---

R52320	USPS First Class Mail will serve as the delivery method for all Checks mailing.
R52330	Mail pick up by USPS First Class mail will occur at Computershare's Bolingbrook (IL) facility at 7:00pm CST, Monday through Friday, excluding NYSE holidays.

5.2.4 Service Level Agreement

R52410	Once live in production, Computershare will not need approval from BMO before printing and mailing any files.
R52420	Files received by 12:00pm CT will be printed and lodged on the same day.
R52425	Files that are received after 12:00pm CT will be printed and lodged on the next business day.
R52430	In the event that a Check needs to be pulled from the mail and returned to BMO or destroyed, BMO will submit a request to their designated Computershare Account Manager and the process will be handled manually.

5.3 Check Image Delivery

5.3.1 Check Image Delivery

R53110	Computershare will create PDF files of check mailpacks for all account numbers flagged in Computershare's database to have check mailpack PDF's produced that were submitted in daily PAIN001 input file(s). The PDF files have the following naming convention: PARTY_ID.CHK_NUM.TRAN_ID.pdf Where the PARTY_ID is extracted from the PAIN001 file, xml node: Document/CstmrCdtTrfInitnt/GrpHdr/InitgPty/Id/OrgId/Othr/Id And CHK_NUM is the check number is extracted from the PAIN001 file, xml node: Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb And the TRAN_ID is extracted from the PAIN001 file, xml node: Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/PmtId/EndToEndId
---------------	--

R53120	The PDF files of the check mailpacks are to be saved into one zip file per Customer ID per production run with the following naming convention: UAT: COMPAYUS.[Customer ID].CHKIMGUST.YYYYMMDD.HHMMSS.PDF.NXXXXXX.zip Prod: COMPAYUS.[Customer ID].CHKIMGUSP.YYYYMMDD.HHMMSS.PDF.NXXXXXX.zip Where: COMPAYUS is hardcoded And where [Customer ID] is extracted from the PAIN001 file, xml node: Document/CstmrCdtTrfInitnt/GrpHdr/InitgPty/Id/OrgId/Othr/Id And where CHKIMGUST [for UAT] or CHKIMGUSP [for Production runs] is hardcoded And where DATE represents the date the zip file is created, and is formatted: MMDDYYYY And where TIMESTAMP is formatted: HHMMSS and reflects the time (in EST) the zip file is generated. And where NXXXXXX = The unique serial number assigned by BMO from the PAIN.001 file
R53130	Once each zip file is generated, it is to be transmitted via sFTP to BMO CIPG.

APPENDIX A - ISO PAIN.001 FILE RECORD STRUCTURE

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.PayeeName	<Cdtr><Nm>	140	Mandatory Payee name will be up to 128 characters, CS should be able to take up to 128 and format from the single <Cdtr><Nm> tag
BR.PayeeAddress	<Cdtr> <PstlAdr> and <BldgNb> and <StrtNm> and <AdrLine>	<Strt><Nm> - 70 <BldgNb> - 16 <AdrLine> - 70	Mandatory Optional Street name is up to 70 Building Number is up to 16 Address Line is up to 70 Street name will appear before building number. Order to be reversed when printed. Address Line is optional Confirm true limit at CS
BR.PayeeCity	<Cdtr> <PstlAdr> <TwnNm>	2-35	Mandatory
BR.PayeeProvince	<Cdtr> <PstlAdr> <CtrySubDvsn>	2-35	Optional Truncate and take the first two letters of the State only.
BR.PayeePostalCode	<Cdtr> <PstlAdr> <PstCd>	3-16	Optional
BR.PayeeCountryCode	<Cdtr> <PstlAdr> <Ctry>	2-2	Mandatory
BR.PayeeCountryName			
BR.MailDestination			

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.Vendor	<Cdtr> <Orgld> <Othr> <Id> OR <Cdtr> <Prvtld> <Othr> <Id>	2-35	Optional One or the other will be present but not both A value of N/A or something similar must be valid for use by clients who do not have a creditor ID
BR. CheckDate	<PmtInf> <ReqdExctnDt>	10	Format will be YYYY-MM-DD per ISO requirements
BR.CheckNumber	<CdtTrfTxInf> <ChqInstr> <ChqNb>	1-35	Confirm true limit at CS
Memo	<MemoFld>		If there is a second occurrence of the <MemoFld>, it will be concatenated to the first occurrence with a space added between the values, creating one field. If there are any additional occurrences of Document/CstmrCdtTrfInitn/PmtInf/CdtTrfTxInf/ChqInstr/MemoFld after a second one, they are to be ignored.
BR.InvoiceDateTable	<RmtInf> <Strd> <RfrdDocInf> <RltdDt>	10-10	Format will be YYYY-MM-DD per ISO requirements
BR.InvoiceNumberTable	<RmtInf> <Strd> <RfrdDocInf> <Nb>	1-35	
BR.InvoiceAmountTable	RmtInf> <Strd> <RfrdDocAmt> <DuePyblAmt Ccy="AAA">	1-18	Optional when the <Strd> record contains a credit note (CREN) in <RmtInf><Strd><RfrdDocInf><Tp><Cd> Optional when If the <Strd> record contains a commercial invoice (CINV) in <RmtInf><Strd><RfrdDocInf><Tp><Cd>

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.DiscountTable	<RmtInf> <Strd> <RfrdDocAmt> <DscntApldAmt Ccy="AAA">	1-18	Optional When CINV commercial invoice is used, this means a reduction in the amount being paid in the remitted amount tag. When CREN credit note is used, this means a reduction to the amount of the credit in the credit note amount tag (ie. There is a \$20 promotion but the credit I'm taking is only \$15.) For either the CINV commercial invoice or the CREN credit note, a "-" sign is hardcoded to the right of the value to show that there is a reduction in the amount being paid in the remitted amount tag.
BR.Additional_Info	<RmtInf> <Strd> < AddtlRmtInf>	1-35	
BR.AmountPaidTable	<RmtInf> <Strd> <RfrdDocAmt> <RmtdAmt Ccy="AAA"> Or <CdtNoteAmtCcy="AAA">	1-18	Either Remitted Amount or Credit Note amount are Mandatory If the <Strd> record contains a credit note (CREN) in <RmtInf><Strd><RfrdDocInf><Tp><Cd> then <RmtdAmtCcy="AAA"> will not be present in the <Strd> record, instead <CdtNoteAmtCcy="AAA"> will be present
BR.Currency	< CdtTrfTxInf > <Amt> <InstdAmt Ccy="AAA">	3-18	Mandatory CAD/USD Currency to be found in the same tag as "amount".
BR.PaymentMethodCode	<PmtInf> <PmtTpInf> <LclInstrm> <Prtry>	3-3	Mandatory 'CHK'=>Check ISO files will not contain EFT transactions

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.GenericMessage	<RmtInf> <Ustrd> And/OR <RmtInf> <Strd> <AddtlRmtInf>	1-140 1-80	Optional <Ustrd> Placement of payment-level freeform text printed under the remittance columns NOT AN OR CONDITION – both should be allowed for any given transaction <AddtlRmtInf> Placement of invoice level freeform text printed under the Invoice date in the first column– Limit of 1 instance per <Strd> record. In the event a client populates the literal NOTE// or DESC// in the <AddtlRmtInf> tag, DO NOT print the literal NOTE// or DESC//. Please suppress these and do not print.
BR.Option3Table	<RmtInf> <Ustrd> <RmtInf> <Strd> <AddtlRmtInf> <RfrdDocInf> <Nb> <DuePyblAmt Ccy="AAA">> <DscntApldAmt Ccy="AAA"> <CdtNoteAmt Ccy="AAA">	1-140 1-80	For Pilgrim's Pride Corp accounts. <Ustrd> tag will store a semi colon delimited data line that will be repeated as needed for all records in the Option 3 table <AddtlRmtInf> tag will store a single instance of a semi colon delimited data line for use in the Option 3 table <Nb> tag will store a single instance of the Quantity Bushels Total <DuePyblAmt> tag will store a single instance of the Gross Amount Total <DscntApldAmt> tag will store a single instance of the Discount Total <CdtNoteAmt> tag will store a single instance of the Net Total
BR.Language	N/A		EN

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.SpecialHandling	<CdtTrfTxInf> <PmtTpInf> <DlvryMtd> <Prtry>	1-35	Optional Values are: Blank or tag not present = USPS to Payee '01' = Courier to Debtor '02' = Courier to Payee '03' = Courier to Client's Special Handling Address
BR.AccountType	<DbtrAcct> <Tp> <Cd>	1-4	Value = CACC to indicate checking account This value is optional and not required at CCS.
BR.TransitNumber	N/A		Member ID will always contain the 9 digit ABA number
BR.Institution	<DbtrAgt> <FinInstnId> <ClrSysMmbld> <Mmbld>	4 -35	Member ID will always contain the 9 digit ABA number
BR.BankTransitNo (2 fields depends on the document type)	N/A		Member ID will always contain the 9 digit ABA number
BR.AccountNumber (2 fields depends on the document type)	<DbtrAcct> <Id> <Othr> <Id>	1-34	Account number up to 12 characters
BR.IssueBank			Else then 'BMO' Should always be BMO, not Harris.
BR.LetterAccountNumber	<CdtTrfTxInf> <PmtId> <EndToEndId>	1-35	The <EndToEndId> is an originator-generated transaction level value that must be unique within the file, and which must be preserved and sent back in pain.002 acknowledgement reporting in phase 1. Ideally all reporting, including recon reporting would include this in the target state (recon reporting may be future phase)
BR.CompanyCode	<Dbtr> <Id> <OrgId> <Othr> <Id>	35	Mandatory this can contain the Computershare company ID ie. 3+DUNS number. The, preferred, location for this value is the <Dbtr><Id><OrgId><Othr><Id> with a trigger value of <SchmeNm><Prtry> of "BMOCOID"

Business Logic Field	Data Specification Fields (BMO)	Field Length	Business Rule (Direct mapping if empty)
BR.DFIQualifier	<DbtrAgt> <FinInstnId> <ClrSysMmbld> <Mmbld>	35	Should always be 'BMO'. Member ID will always contain the 9 digit ABA number
BR.CheckAmount	<CdtTrfTxInf> <Amt> <InstdAmt Ccy=""AAA"">	1/18	The instructed amount must equal the total of all <Strd> record Remitted amounts MINUS the total of all <Strd> Credit Note amounts.
BR.ReferenceId	<CdtTrfTxInf> <PmtId> <InstrId>	1-35	If this is a batch level ID, then the <InstrId> should be referenced. If present, the <InstrId> should also be passed along to the pain.002 acknowledgement. It is an optional field.

APPENDIX B - ISO PAIN.002 FILE RECORD STRUCTURE

Field	Mapping from BMO Pain.001 ISO input file	Field Length	Business Rules
Customer Payment Status Report <CstmrPmtStsRpt>			Parent Element.
GroupHeader <GrpHdr>			Parent Element
Message Identification <Msdl>	<GrpHdr> <Msgld>	35	Use a combination of date/time stamp as a unique identifier.
Creation Date Time <CreDtTm>	<GrpHdr> <CreDtTm>	30	<i>Mapped from Pain.001</i>
Initiating Party <InitgPty>			Parent Element
Name <Nm>		140	Hardcoded "BMO"
Identification <Id>		35	Hardcoded "BOFMCAM2"
Scheme Name <SchmeNm>			Parent Element
Code <Cd>		4	Hardcoded "BANK"
ContactDetails <CtctDtls>			Parent Element
Name <Nm>		140	Hardcoded "BMO Harris TPS Helpdesk"
Email Address <EmailAdr>		140	Hardcoded TPSCAD.HelpDesk@bmo.com
oneNumber <PhneNb>		33	Hardcoded "1-800-565-6444"

Field	Mapping from BMO Pain.001 ISO input file	Field Length	Business Rules
Original Group Information And Status <OrgnlGrpInfAndSts>			Parent Element
Original Message Identification <OrgnlMsgId>	<GrpHdr> <MsgId>	35	<i>Mapped from Pain.001</i>
Original Message Name Identification <OrgnlMsgNmId>		35	Hardcoded "pain.001.001.03"
Original Number Of Transactions <OrgnlNbOfTx>		15	Number of individual transactions contained in the original message.
Original Control Sum <OrgnlCtrlSum>		18	Total of all individual amounts included in the original message, irrespective of currencies.
Group Status <GrpSts>		4	If the entire file is accepted "ACCP" will appear, if the entire file is rejected, "RJCT" will appear, if part of the file is accepted and part is rejected, "PART" will appear
Number Of Transactions Per Status <NbOfTxPerSts>			Parent Element
Detailed Number Of Transactions <DtldNbOfTx>		15	Count of transactions per status For ACCP: QTY_02_380(55) - Net to Pay item count For RJCT: QTY_02_380(54) - Bank Reject item count
Detailed Status <DtldSts>		4	Status code per status
Detailed Control Sum <DtldCtrlSum>		18	Amount per status
Original Payment Information And Status <OrgnlPmtInfAndSts>			Parent Element
Original Payment Information Identification <OrgnlPmtInfId>	<PmtInf> <PmtInfId>	35	<i>Mapped from Pain.001</i>
Original Number Of Transactions <OrgnlNbOfTx>		15	Total number of Number of field <TxInfAndSts> Below

Field	Mapping from BMO Pain.001 ISO input file	Field Length	Business Rules
Additional Information <AddtlInf>		105	Up to Two Occurrences: 1) Concatenation of "FINALAMT/" and <InstdAmt> from pain.001/pain.008 2) Concatenation of "CONTRAAMT/" and (Rejected) <InstdAmt> from pain.001/pain.008
Transaction Information And Status <TxInfAndSts>			Parent Element
Status Identification <StsId>		35	Optional Can be used by Computershare if there is some sort of reference ID assigned when a transaction is Accepted or Rejected
Original Instruction Identification <OrgnlInstrId>	<PmtInf> <CdtTrfTxInf> <PmtId> <InstrId>	35	<i>Mapped from Pain.001</i>
Original End To End Identification <OrgnlEndToEndId>	<PmtInf> <CdtTrfTxInf> <PmtId> <EndToEndId>	35	<i>Mapped from Pain.001</i>
Transaction Status <TxSts>		4	Can be either ACCP or RJCT
Status Reason Information <StsRsnInf>			Parent Element Only used for rejected transactions
Reason <Rsn>			Parent Element
Code <Cd>		4	If <TxSts> = "RJCT" then Value = "NARR" OR Reason Code as per Status Reason Codes for Canada attachment. If the transaction status <TxSts> is "RJCT" then the literal "NARR" will appear to advise you to look at <AddtlInf> for details about the reject
<AddtlInf>			When NARR or any reason is used, this must be present and populated To contain the reason for the status of RJCT.

Field	Mapping from BMO Pain.001 ISO input file	Field Length	Business Rules
Original Transaction Reference <OrgnITxRef>			Parent Element
Amount <Amt>			Parent Element
Instructed Amount <InstdAmt Ccy="AAA">	<PmtInf> <CdtTrfTxInf> <Amt> <InstdAmt Ccy="AAA">	18	Mapped from Pain.001
Requested Execution Date <ReqdExctnDt>	<PmtInf> <ReqdExctnDt>	30	Mapped from Pain.001

APPENDIX C - GLOSSARY OF TERMS

Table of - Glossary of Terms

Term	Description
Run/Stream	A group of records that are sent out together from the data file creating a mailpack.
Set	A group of records required creating one mailpack.
Errors	This refers to data dependent problems.
Extracts	These are accounts that <Client Name> have requested not to be printed or mail-lodged.
Returns	These are accounts that <Client Name> have requested to be printed and returned to them.
Simplex	Single-sided laser printing.
Duplex	Double-sided laser printing.
Multiplex	Mixed simplex and duplex laser printing.

Appendix D - REJECT REASON CODES

File Level Validations

Business Condition	Action	Reason Code	Definition of Reason Code
UTF – 8 encoding not detected	Fail the entire file	NARR	Data field '\$tagPathString.'" is not encoded using UTF8; file is not encoded in UTF-8 XML format
Schema Validation as per ISO 20022 standard	Fail the entire file	TD02	File format not readable
XML file is empty	Fail the entire file	TD03	Incorrect file structure
File contains zero transactions	Fail the entire file	TD03	Incorrect file structure
File is not a valid XML extension file	Fail the entire file	TD02	File format not readable
File name does not conform to the agreed upon format	Fail the entire file	TD02	File format not readable
Duplicate file validation	Fail the entire file	DU01	The data file is an exact duplicate of one of the previously processed files.
Control Sum Validation	Fail the entire file	AM10	The total amount not does not match total of cheques processed.

Payment Info Group Level Validations

Business Condition	Action	Reason Code	Definition of Reason Code	BMO XPATH(S)
Payer Name is missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Payer Detail Field is Missing: Payer Name, Data mapping at right from Payer Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/Dbtr/Nm

Payer Street Name is missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Payer Detail Field is Missing: Payer Street Name Data mapping at right from Payer Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/Dbtr/PstlAdr/StrtNm
Payer City is missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Payer Detail Field is Missing: City Data mapping at right from Payer Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/Dbtr/PstlAdr/TwnNm
Payer State is missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Payer Detail Field is Missing: State Data mapping at right from Payer Details in Appendix A	Document/CstmrCdtTrfIntr/PmtInf/Dbtr/PstlAdr/CtrySubDvsn
Payer Zip is missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Payer Detail Field is Missing: Zip Data mapping at right from Payer Details in Appendix A	Document/CstmrCdtTrfIntr/PmtInf/Dbtr/PstlAdr/PstCd
Account Number is Missing	Reject all transactions in the Payment	CH21	Mandatory Check Detail Field is Missing: Account Number	Document/CstmrCdtTrfIntr/PmtInf/DbtrAcct/Id/Othr/Id

	Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups		OR Account Number is not found in CCS's Encompass2 database	
Routing Number is Missing	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	CH21	Mandatory Check Detail Field is Missing: Routing Number	Document/CstmrCdtTrfIntr/PmtInf/DbtrAgt/FinInstnId/CtrSysMmbld/Mmbld
Transit [Routing] Numbers/ MICR Details invalid	Reject all transactions in the Payment Info Group with the same Reason Code and Additional Info, and continue processing the rest of the Payment Info Groups	NARR	Routing number is not found in our routing number database nor in CCS's Encompass2 database for the Account Number passed to us in the Pain001 file.	Document/CstmrCdtTrfIntr/PmtInf/DbtrAgt/FinInstnId/CtrSysMmbld/Mmbld

Transaction Level Validations

Business Condition	Action	Reason Code	Definition of Reason Code	BMO XPATH(S)
--------------------	--------	-------------	---------------------------	--------------

Payee Name is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Payee Detail Field is Missing: Payee Name, Data mapping at right from Check Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/Nm
Payee Street Name or Address Line is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Payee Detail Field is Missing: Street Name AND Address Line Data mapping at right from Check Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/PstlAdr/StrtNm /Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/PstlAdr/AdrLine
Payee City is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Payee Detail Field is Missing: City Data mapping at right from Check Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/PstlAdr/TwnNm
Payer State is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Payee Detail Field is Missing: State Data mapping at right from Check Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/PstlAdr/CtrySubDvsn
Payer Zip is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Payee Detail Field is Missing: ZIP Data mapping at right from Check Details in Appendix A	/Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/Cdtr/PstlAdr/PstCd

Check Number is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Check Detail Field is Missing: Check Amount, Check Date; Data mapping at right from Check Details in Appendix A	Document/CstmrCdtTrfInitt/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb
Check Date is missing	Skip the record, continue processing the rest of the file	CH21	Mandatory Check Detail Field is Missing: Check Date Data mapping at right from Check Details in Appendix A	Document/CstmrCdtTrfInitt/PmtInf/ReqdExctnDt
Cheque Number is above 10 digits	Skip the record, continue processing the rest of the file	NARR	Data exceeds the maximum character limit required.	Document/CstmrCdtTrfInitt/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb
Cheque Number is not numeric	Skip the record, continue processing the rest of the file	FF09	If the cheque number is not numeric, an error is produced.	Document/CstmrCdtTrfInitt/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb
Memo Field - Max 35 characters per field; 2 field occurrences allowed	Skip the record, continue processing the rest of the file	NARR	Data exceeds the maximum character limit required for one or both field occurrences.	Document/CstmrCdtTrfInitt/PmtInf/CdtTrfTxInf/ChqInstr/MemoFld
Cheque Amount - must be greater than 0 USD	Skip the record, continue processing	AM12	If the cheque amount equals 0 or a negative value, this error is produced.	Document/CstmrCdtTrfInitt/PmtInf/CdtTrfTxInf/Amt/InstdAmt

	the rest of the file			
Currency (amount in USD)	Skip the record, continue processing the rest of the file	CURR	Currency of the payment is not in USD	N/A
Duplicate check found within the input file	Skip the record, continue processing the rest of the file	DUPL	If a duplicate cheque number is passed, this error will be generated.	Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb
Duplicate cheque found: '\$set->{BR.Cheque Number}' was already processed (within last year) in data file	Skip the record, continue processing the rest of the file	DUPL	If a duplicate cheque number is passed, this error will be generated.	Document/CstmrCdtTrfIntr/PmtInf/CdtTrfTxInf/ChqInstr/ChqNb